**Constitution of matter**

The constitution of matter can simply be understood as the building block of matter be it solid (salt, sugar) liquid (water, petrol) or gas (air, diesel, water vapour) or it can be defined as the basic unit of matter.

## STATES OF MATTER

Matter can exist in three distinct states which are; solid liquid and gas example;

Water can exist in solid state as ice, liquid state as ordinary water and gaseous state as water vapour.

When heat is applied to water in liquid state there will be rise in temperature (hotness of the water) until it reaches 100°C (boiling point of water) water vapour will begin to form which is the gaseous state of water.

The process by which water changes from liquid to gaseous state is known as vaporization.

**NOTE:** that not all matter can exist in these three states (solid, liquid and gas) because some can change directly from solid to liquid state without passing through the liquid state a phenomenon referred to as sublimation

## CHEMICAL EQUILIBRIUM

In chemical equilibrium of a reversible reaction, the rate of the formation of the product is equal to the rate of the formation of the reactant which is result to no net changes in the concentration of the reactants and products.

## FACTORS AFFECTING EQUILIBRIUM

The factors that affect equilibrium are:

i. Temperature
ii. Concentration

iii. Pressure or volume of gas.

iv. Catalyst does not affect equilibrium but lower the activation energy for faster attainment of equilibrium.

## REVERSIBLE REACTION

Reversible reactions are reactions that can proceeds from left to right and vice-versa.

*Example of reversible of reactions are.*

$$H_2 + I_{2\,(g)} \rightleftharpoons 2HI_{(g)}$$

$$N_2 + 3H_{2\,(g)} \rightleftharpoons 2NH_{3\,(g)}$$

## LECHATELIERS PRINCIPLE

Lechateliers principles state that if a system is in equilibrium and external constraint such as change in concentration, pressure or volume is imposed. The equilibrium will shaft to reduce or counteract the imposed constraint.

**Equilibrium Constant Kc:**

A number that expresses the relationship between the amounts of products and reactants present at equilibrium in a reversible chemical reaction at a given temperature.

For instance

$$A + B \rightleftharpoons C + D$$

A mixture of products and reactants in a state of chemical equilibrium is known as an equilibrium mixture.

$$Kc = \frac{\{C\}\{D\}}{\{A\}\{B\}}$$

Representation of the Equilibrium Constant

For a balanced reaction of the type, $aA + bB \rightleftharpoons cC + dD$

According to the law of mass action, the constant value obtained by relating equilibrium concentrations of reactants and products is called the equilibrium constant.

For the forward reaction, this is given by

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

The activities of solids, pure liquids, and solvents are not approximated with their molarities. Instead these activities are defined to have a value equal to 1 (one)

Example

$$HF_{(aq)} + H_2O_{(l)} \rightleftharpoons H_3O^+_{(aq)} + F^-_{(aq)}$$

$$K_c = \frac{a_{H_3O^+} \cdot a_{F^-}}{a_{HF} \cdot a_{H_2O}} \approx \frac{[H_3O^+][F^-]}{[HF](1)} = \frac{[H_3O^+][F^-]}{[HF]}$$

The equilibrium constant for the reverse reaction is the inverse of the forward reaction and is given by:

$$K'_c = \frac{1}{K_c} = \frac{[A]^a[B]^b}{[C]^c[D]^d}$$

*The lower case letters in the balanced equation represent the number of moles of each substance, the upper case letters represent the substance itself.

- If K>1 then equilibrium favors products
- If K<1 then equilibrium favors the reactants

Equilibrium Constant of Pressure

Gaseous reaction equilibria are often expressed in terms of partial pressures. The equilibrium constant of pressure gives the ratio of pressure of products over reactants for a reaction that is at equilibrium (again, the pressures of all species are raised to the powers of their respective coefficients). The equilibrium constant is written as Kp, as shown for the reaction:

$$aA_{(g)} + bB_{(g)} \rightleftharpoons gG_{(g)} + hH_{(g)}$$

$$K_p = \frac{p^g G \, p^h H}{p^a A \, p^b B}$$

- Where p can have units of pressure (e.g., atm or bar).

**Homogeneous Reactions**

A homogeneous reaction is one where the <u>states of matter</u> of the products and reactions are all the same (the word "homo" means "same").

Example

The synthesis of methanol from a carbon monoxide-hydrogen mixture is *gaseous* homogeneous mixture

$$CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$$

Example two

The synthesis of ammonia is another example of a *gaseous* homogeneous mixture:

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

Heterogeneous Reactions

A heterogeneous reaction is one in which one or more states within the reaction differ a heterogeneous mixture dealing with particles in both the *solid* and *aqueous* states:.

Example 1,

The formation of an aqueous solution of lead(II) iodide creates

$$PbI_{2(s)} \rightleftharpoons Pb^{+2}_{(aq)} + 2I^{-}_{(aq)}$$

Example 2

The decomposition of sodium hydrogen carbonate (baking soda) at high elevations this reaction deals with molecules in both the *solid* and *gaseous* states:

$$2NaHCO_{3(s)} \rightleftharpoons Na_2CO_{3(s)} + H_2O_{(g)} + CO_{2(g)}$$

Example 3: The reaction of Carbon and Oxygen

$$C_{(s)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$$

ASSIGNMENT

1. Write the equilibrium constant expression for each reaction.

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

$$Kc = \frac{[SO_3]^2}{[O_2][SO_2]^2}$$

$N_2O_{(g)} + \frac{1}{2} O_{2(g)} \rightleftharpoons 2NO_{(g)}$

$$Kc = \frac{[NO]^2}{[O_2]^{0.5}[N_2O]}$$

$Cu_{(s)} + 2Ag^{+}_{(aq)} \rightleftharpoons Cu^{+2}_{(aq)} + 2Ag_{(s)}$

$$Kc = \frac{[Cu^{+2}]}{[Ag^{+}]^2}$$

$CaCO_{3(g)} \rightleftharpoons CaCO_{(s)} + CO_{2(g)}$

$$Kc = \frac{[CO_2]}{[CaCO_3]}$$

2. What is the Kc of the following reaction?

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

With concentration $SO_{2(g)} = 0.2M$ $O_{2(g)} = 0.5M$ $SO_{3(g)} = 0.7M$

Answer is Kc: 24.5

3. For the same reaction, the differing concentrations:

$SO_{2(g)} = 0.1M$ $O_{2(g)} = 0.3M$ $SO_{3(g)} = 0.5M$

Qc : 83.33

$$Kc = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{[0.5]^2}{[0.1]^2[0.3]} = \frac{0.25}{0.01 \times 0.3} = \frac{0.25}{0.003} = 83.33$$

4. Write the Partial Pressure Equilibrium:

$C_{(s)}+O_{2(g)} \rightarrow CO_{2(g)}$

$$Kp = \frac{PCO_2}{PO_2}$$

5. Write the chemical reaction for the following equilibrium constant:

$$Kp = \frac{P^2HI}{PH_2 \times PI_2}$$

## MASS ACTION LAW

Mass action law states that a chemical reaction frequency is proportional to the active masses of reacting materials at a constant temperature. Kc is the reaction constant of equilibrium, while Kp is the constant of equilibrium found by applying partial pressure.

### Oxidation reduction reaction (Redox)

Oxidation reactions:- This reaction can be defined based on the following processes

**i. Gaining of oxygen**

The term oxidation was initially used to described reactions in which oxygen combined with an element or a compound to form another substance. A substance is oxidized if it gains oxygen e.g.

$2Ca_{(s)} + O_{2(g)} \rightarrow 2CaO_{(s)}$.

Since calcium has gained oxygen, then it is said to have been oxidized to calcium oxide and the process is called oxidation

**ii. Removing hydrogen**

A substance is oxidized if it loses hydrogen. When ammonia gas is passed over heated cupper (II) oxide, the following reaction occurs

$2NH_{3(g)} + 3CuO_{(s)} \rightarrow N_{2(g)} + 3Cu_{(s)} + 3H_2O_{(l)}$

Ammonia has lost hydrogen; it has been oxidized to nitrogen

**iii. Adding electronegative elements**

Oxidization can be also be defined as the addition of electronegative elements e.g.

$Cu_{(s)} + Cl_{2(g)} \rightarrow CuCl_{2(s)}$.

Since the chlorine has been added to the cupper to form cupper (ii) chlorine, you say that cupper has been oxidized

**iv. Losing electrons**

Oxidation can be defined as lose of electrons from a substance e.g. in $Mg_{(s)} + Cl_{2(g)} \rightarrow MgCl_2$

- Looking at the above reaction, oxidation has taken place because electrons transfer has taken place. The transfer of electrons can be explained by the following half-equations

$Mg_{(s)} \rightarrow Mg^{2+}_{(s)} + 2e^-$

$Cl_{2(g)} + 2e^- \rightarrow 2Cl_{(s)}$

**v. Increase of oxidation state**

The oxidation state (oxidation number) is the charge on atom of an element would have if it existed as an ion in a compound.

Example consider the reaction between zinc and a solution of cupper (II) ions the zinc atom, Zn (oxidation state = 0) becomes the zinc ion, $Zn^{2+}$ (oxidation state = +2). The oxidation state of the zinc atom has increased. When this happens, Zinc is said to be oxidized.

Therefore, if the oxidation state of a substance increases, the substance is said to be oxidized.

Oxidation

0 → +2

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

## REDUCTION REACTIONS

**i. Removing oxygen**

Reduction can be defined as the remove of oxygen from a substance (a reverse of oxidation). E.g.

$$Zn_{(s)} + CuO_{(s)} \rightarrow ZnO_{(s)} + Cu_{(s)}$$

In this reaction, the cupper (II) oxide has lost its oxygen. It is said to have been reduced to metallic cupper.

**ii. Gaining hydrogen**

Reduction can be defined as gaining of hydrogen e.g.

$H_{2(g)} + CL2_{(g)} \rightarrow 2HCL_{(g)}$

Chlorine has gained hydrogen, so it has been reduced.

**iii. Adding electropositive elements**

Electropositive elements have a tendency to lose electrons to lose electrons to form positive ions. These elements are usually metal. Group I metals are especially highly electropositive.

Consider the reaction between sodium and chlorine

$Na_{(s)} + Cl_{2(g)} \rightarrow 2NaCl_{(s)}$

Sodium which is electropositive is added to chlorine to form sodium chlorine. Thus, the chlorine has been reduced to its chloride.

**iv. Gaining electrons**

Reduction is also defined as the addition of electron to substance. A substance is reduced is reduced if it gains electrons during a reaction e.g.

$2FeCl_{3(aq)} + H2S_{(g)} \rightarrow 2FeCL_{2(aq)} + 2HCl_{(aq)} + S_{(s)}$

The above chemical equation can be rewriten as an ionic equation

$2Fe^{3+}_{(aq)} + H_2S_{(g)} \rightarrow 2Fe^{3+}_{(aq)} + 2H^+_{(aq)} + S_{(s)}$

In this reaction. Iron (III) ions have gained electrons to form iron (II) ions. This means that iron (III) ions have been reduced to iron (II) ions

$$2Fe^{3+}_{(aq)} + 2e^- \rightarrow 2Fe^{2+}_{(aq)}$$

**vi. Decreasing the oxidation state.**

Reduction can be defined as the decrease in the oxidation state of a substance after a reaction e.g.

$Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

The cupper (II) ion, $Cu^{2+}$ (oxidation state = +2), becomes the cupper atom, Cu oxidation state = 0). The oxidation state of the $Cu^{2+}$ iron has decreased, when this happens, the $Cu^{2+}$ ion to be reduced to cupper.

**Halogens**

Examples of the halogens that are good oxidizing agents include iodine, bromine, chlorine, and fluorine. Fluorine is said to be the strongest elemental oxidizing agent due to its highest electronegativity,

Many other oxidizing agents are commonly used industrially as well as in the day-to-day lives of humans.

1. Potassium Nitrate ($KNO_3$),
2. Nitric acid ($HNO_3$),
3. Perchloric acid ($HClO_4$),
4. Sulfuric acid ($H_2SO_4$).
5. Diatomic oxygen ($O_2$),
6. Diatomic chlorine ($Cl_2$),
7. Ozone ($O_3$).
8. Hydrogen peroxide ($H_2O_2$)

A reducing agent is a substance that causes reduction by losing electrons; therefore it gets oxidized.

Reducing agents include :

| | | |
|---|---|---|
| 1. Chlorine ion | - | $Cl^-$, |
| 2. Cupper | - | Cu, |
| 3. Hydrogen Ion | - | $H^-$ |
| 4. Hafnium | - | HF, |
| 5. Lead | - | Pb, |
| 6. Zinc | - | Zn |

**Postulates of the kinetic theory of gases**

The kinetic theory of gases assumes that gases are made up of identical molecules that:

i. Are very widely separated

ii. Have negligible volume (i.e. zero volume)

iii. Exert no forces of attraction on one another

iv. Move continuously and randomly in straight lines

v. Have perfectly elastic collision

vi. Have an average kinetic energy which is directly proportional to the absolute temperature

**What is a mean free path:** The mean free path is the average distance traveled by a moving molecule between collisions.

**Collision number:** The number of collisions suffered by a single molecule per unit time per unit volume of the gas.

**Collision Diameter:** The following figure shows the collision of two molecules. Two such molecules approach each other as a result of attractive forces present in these molecules.



060